DBMS – PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

Project Title: Helios Watch Store E-Commerce Database

Skill Task: DBMS

Week: 6

1. Database Design

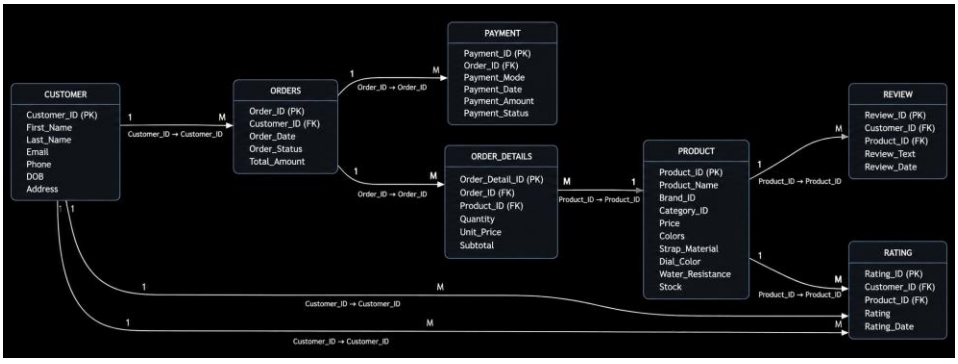
1.1 Review Table

Attribute	Data Type	Constraint
Review_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Review_Text	VARCHAR2(500)	NOT NULL
Review_Date	DATE	NOT NULL

1.2 Rating Table

Attribute	Data Type	Constraint
Rating_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Rating	NUMBER	NOT NULL
Rating_Date	DATE	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Review Table

```
SQL> CREATE TABLE Review (
  2      Review_ID     NUMBER PRIMARY KEY,
  3      Customer_ID   NUMBER NOT NULL,
  4      Product_ID    NUMBER NOT NULL,
  5      Review_Text   VARCHAR2(500) NOT NULL,
  6      Review_Date   DATE NOT NULL,
  7      CONSTRAINT FK_Review_Customer
  8          FOREIGN KEY (Customer_ID)
  9          REFERENCES Customer(Customer_ID),
 10      CONSTRAINT FK_Review_Product
 11          FOREIGN KEY (Product_ID)
 12          REFERENCES Product(Product_ID)
 13  );

Table created.
```

3.2 Create Rating Table

```
SQL> CREATE TABLE Rating (
  2      Rating_ID     NUMBER PRIMARY KEY,
  3      Customer_ID   NUMBER NOT NULL,
  4      Product_ID    NUMBER NOT NULL,
  5      Rating        NUMBER NOT NULL,
  6      Rating_Date   DATE NOT NULL,
  7      CONSTRAINT FK_Rating_Customer
  8          FOREIGN KEY (Customer_ID)
  9          REFERENCES Customer(Customer_ID),
 10      CONSTRAINT FK_Rating_Product
 11          FOREIGN KEY (Product_ID)
 12          REFERENCES Product(Product_ID)
 13  );

Table created.
```

4. Review Data Insertion

```
SQL> INSERT INTO Review
  2  VALUES (601, 1, 101, 'Good quality and elegant design.', TO_DATE('10-09-2026','DD-MM-YYYY'));


1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (602, 2, 102, 'Beautiful watch with premium finish.', TO_DATE('11-09-2026','DD-MM-YYYY
'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (603, 3, 103, 'Good sports watch and easy to use.', TO_DATE('12-09-2026','DD-MM-YYYY')
);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (604, 4, 104, 'Useful smart features and good display.', TO_DATE('13-09-2026','DD-MM-Y
YYY'));

1 row created.

SQL>
```

```
SQL> INSERT INTO Review
  2  VALUES (605, 5, 105, 'Comfortable and suitable for daily use.', TO_DATE('14-09-2026','DD-MM-Y
YYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (606, 6, 106, 'Strong design and excellent performance.', TO_DATE('15-09-2026','DD-MM-
YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (607, 7, 107, 'Great outdoor watch with good water resistance.', TO_DATE('16-09-2026',
'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (608, 8, 108, 'Premium looking watch with excellent finish.', TO_DATE('17-09-2026','DD
-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (609, 9, 101, 'Classic design and good value for money.', TO_DATE('18-09-2026','DD-MM-
```

5. Rating Data Insertion

```
SQL> INSERT INTO Rating
  2  VALUES (702, 2, 102, 5, TO_DATE('11-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (703, 3, 103, 4, TO_DATE('12-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (704, 4, 104, 4, TO_DATE('13-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (705, 5, 105, 5, TO_DATE('14-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (706, 6, 106, 5, TO_DATE('15-09-2026','DD-MM-YYYY'));

1 row created.
```

```
SQL>
SQL> INSERT INTO Rating
  2  VALUES (707, 7, 107, 4, TO_DATE('16-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (708, 8, 108, 5, TO_DATE('17-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2  VALUES (709, 9, 101, 4, TO_DATE('18-09-2026','DD-MM-YYYY'));

1 row created.
```

6. Retrieve Product Review Details

```
SQL> SELECT
  2      r.Review_ID,
  3      c.Customer_ID,
  4      c.First_Name,
  5      c.Last_Name,
  6      p.Product_ID,
  7      p.Product_Name,
  8      r.Review_Text,
  9      r.Review_Date
 10  FROM Review r
 11  JOIN Customer c
 12      ON r.Customer_ID = c.Customer_ID
 13  JOIN Product p
 14      ON r.Product_ID = p.Product_ID
 15  ORDER BY r.Review_ID;

 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       601           1 Renu
Srinivasan                                             101
Classic Silver Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Good quality and elegant design.
10-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       602           2 Arjun
Kumar                                              102
Elegant Gold Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Beautiful watch with premium finish.
11-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       603           3 Priya
Kumar                                                   103
Digital Sport Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Good sports watch and easy to use.
12-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       604           4 Rahul
Sharma                                                  104
Smart Fit Pro
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Useful smart features and good display.
13-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       605           5 Keerthana
Mohan                                                  105
Smart Active Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Comfortable and suitable for daily use.
14-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- --------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       606           6 Nandhini
Ravi                                                   106
Sports Chronograph
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ----------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Strong design and excellent performance.
15-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ----------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       607           7 Karthik
Prakash                                              107
Outdoor Pro Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ----------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Great outdoor watch with good water resistance.
16-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ----------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       608           8 Swetha
Balaji                                               108
Royal Gold Edition
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ------------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Premium looking watch with excellent finish.
17-SEP-26


 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ------------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
       609           9 Vignesh
Ganesh                                                  101
Classic Silver Watch
```

```
 REVIEW_ID CUSTOMER_ID FIRST_NAME
---------- ----------- ------------------------------------------------------
LAST_NAME                                          PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_TEXT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Classic design and good value for money.
18-SEP-26


9 rows selected.
```

7. Calculate Average Product Ratings

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Product p
  6  JOIN Rating r
  7      ON p.Product_ID = r.Product_ID
  8  GROUP BY
  9      p.Product_ID,
 10      p.Product_Name
 11  ORDER BY Average_Rating DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
       102
Elegant Gold Watch
             5

       106
Sports Chronograph
             5
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------

       108
Royal Gold Edition
             5

       105
Smart Active Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
             5

       101
Classic Silver Watch
           4.5

       107
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
AVERAGE_RATING
--------------
Outdoor Pro Watch
             4

       103
Digital Sport Watch
             4


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
AVERAGE_RATING
--------------
       104
Smart Fit Pro
             4


8 rows selected.
```

8. Identify Highly Rated Products

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      ROUND(AVG(r.Rating), 2) AS Average_Rating
  5  FROM Product p
  6  JOIN Rating r
  7      ON p.Product_ID = r.Product_ID
  8  GROUP BY
  9      p.Product_ID,
 10      p.Product_Name
 11  HAVING AVG(r.Rating) >= 4
 12  ORDER BY Average_Rating DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
AVERAGE_RATING
--------------
       102
Elegant Gold Watch
             5

       106
Sports Chronograph
             5
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
AVERAGE_RATING
--------------

       108
Royal Gold Edition
             5

       105
Smart Active Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
AVERAGE_RATING
--------------
             5

       101
Classic Silver Watch
           4.5

       107
```

```
 PRODUCT_ID
 ----------
 PRODUCT_NAME
 --------------------------------------------------------------
 AVERAGE_RATING
 --------------
 Outdoor Pro Watch
              4

        103
 Digital Sport Watch
              4


 PRODUCT_ID
 ----------
 PRODUCT_NAME
 --------------------------------------------------------------
 AVERAGE_RATING
 --------------
        104
 Smart Fit Pro
              4


 8 rows selected.
```

9. Product Rating Report

```
SQL> SELECT
  2      r.Rating_ID,
  3      p.Product_ID,
  4      p.Product_Name,
  5      r.Customer_ID,
  6      r.Rating,
  7      r.Rating_Date
  8  FROM Rating r
  9  JOIN Product p
 10      ON r.Product_ID = p.Product_ID
 11  ORDER BY r.Rating_ID;

 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------
       701        101
Classic Silver Watch
          1          5 10-SEP-26

       702        102
Elegant Gold Watch
          2          5 11-SEP-26
```

```
 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------

       703        103
Digital Sport Watch
          3          4 12-SEP-26

       704        104
Smart Fit Pro

 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------
          4          4 13-SEP-26

       705        105
Smart Active Watch
          5          5 14-SEP-26

       706        106
```

```
 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------
Sports Chronograph
          6          5 15-SEP-26

       707        107
Outdoor Pro Watch
          7          4 16-SEP-26


 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------
       708        108
Royal Gold Edition
          8          5 17-SEP-26

       709        101
Classic Silver Watch
          9          4 18-SEP-26
```

```
 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------
      708        108
Royal Gold Edition
         8          5 17-SEP-26

      709        101
Classic Silver Watch
         9          4 18-SEP-26

 RATING_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CUSTOMER_ID     RATING RATING_DA
----------- ---------- ---------


9 rows selected.
```